

Software Engineering

How To Create Software Products Everyone Loves!

Module 11

UI/UX Fundamentals: Design Systems

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What Is a Design System?

A **design system** is a collection of **methods** and **processes** that make the **teams building digital products collaborate in an organized manner.**

Base values

Components

Code snippets

Definitions

Documentation

Atomic Design Approach

Its premise is that starting small (with atoms, or base values), we define more and more complex structures, ending up with the final page layouts



WHY?

The main advantages of a design system are much higher levels of **consistency** and **faster**, more **precise implementations**.

The **team** can **focus on actually building innovation** without spending time figuring out which pieces to use for building a popup window

INTERNAL CONSISTENCY

It means the same kind of object within the product should always look and work the same way

E-mail

Please enter your e-mail...

E-mail

Please enter your e-mail...

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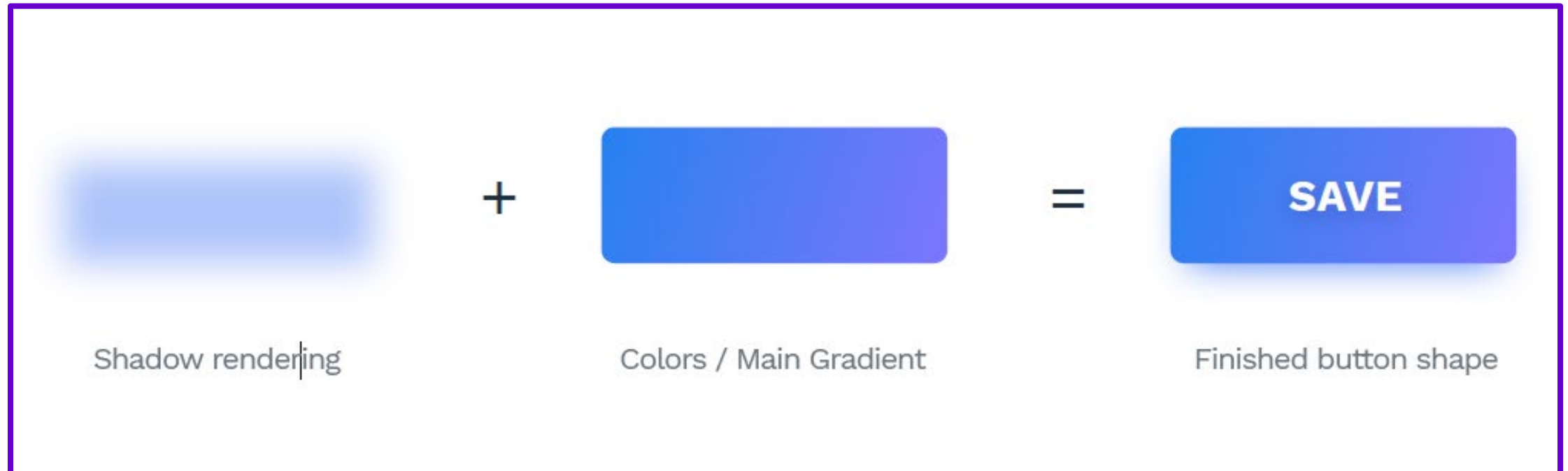
WHERE TO START?

When building a design system for an **existing brand**, you **need to start by analyzing the brand book** and **existing digital products the company has**.

The things you need to define first are **colors, fonts**, and the **grid**. They are the top-level qualities of our design library.

WHERE TO START?

A button is a combination of shape, color, margins, and a font. Split it into its individual components and document each one separately.



Shadow / CTA : Color, blur value, position.

Define your colors first so that they can merge with the shadow description as well.

COLORS/1 HYPE4MAIN



COLORS/3 PINK



COLORS/4 DARK GREY



COLORS/5 GREY



COLORS/6 LIGHT



COLORS/7 WHITE



NESTING AND SHARED PROPERTIES

When you define your color library and the ways they can be used (which color goes where), you can start assigning them to objects.



CSS



Button Shadow

```
box-shadow: 0 4px 9px  
-4px #8C6EE8;
```



Colors / Main Gradient

```
background-image: linear-  
gradient(-49deg, #8E74FF  
0%, #2781F0 100%);  
border-radius: 3px;
```



Global rules

```
font-family: Multi-Black;  
font-size: 10px;  
color: #FFFFFF;  
letter-spacing: 0.15px;  
text-align: center;  
text-shadow: 0 2px 4px  
rgba(0,0,0,0.12);  
line-height: 44px;  
padding: 0 24px;
```

COMMON PARTS

The next step is defining the common code parts

If your button line-height is always 44 points, you can call it "Line-height/button/ standard" in your documentation

If all the code documentation uses are pre-set numbers from a single source (design tokens), you can have a lot more control of the core of your Design System.

DESIGN TOKEN

A design token is an overriding value that you define in one place and then use its token everywhere else. It allows for faster global changes in the future.

Token	Size
<code>\$grid-base</code>	8px
<code>\$grid-small</code>	16px
<code>\$grid-medium</code>	24px
<code>\$grid-large</code>	32px
<code>\$grid-xlarge</code>	48px

Component Definitions

Component definitions (for example, a button's size, spacing, margins, and line-height) should use Design Tokens whenever possible.



Properties

line-height: \$grid-xlarge
height: \$grid-xlarge

Margins

padding: 0 \$grid-medium;
margin: \$grid-small

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SOURCE OF TRUTH

Most design systems start as a component/symbol library in your favorite design tool.

To make your system genuinely work in your company, you need to document it. That **documentation** becomes your "Source of Truth."

You should update it regularly, as design systems are never really complete.

HOW TO DOCUMENT?

What's the update process (who and how accepts a change?)

What's the format and the language used (has to match the company)

Who is responsible for improving the Design System

THE WIKI

Your documentation should be a structured wiki with **versioning, change tracking**, and notes.

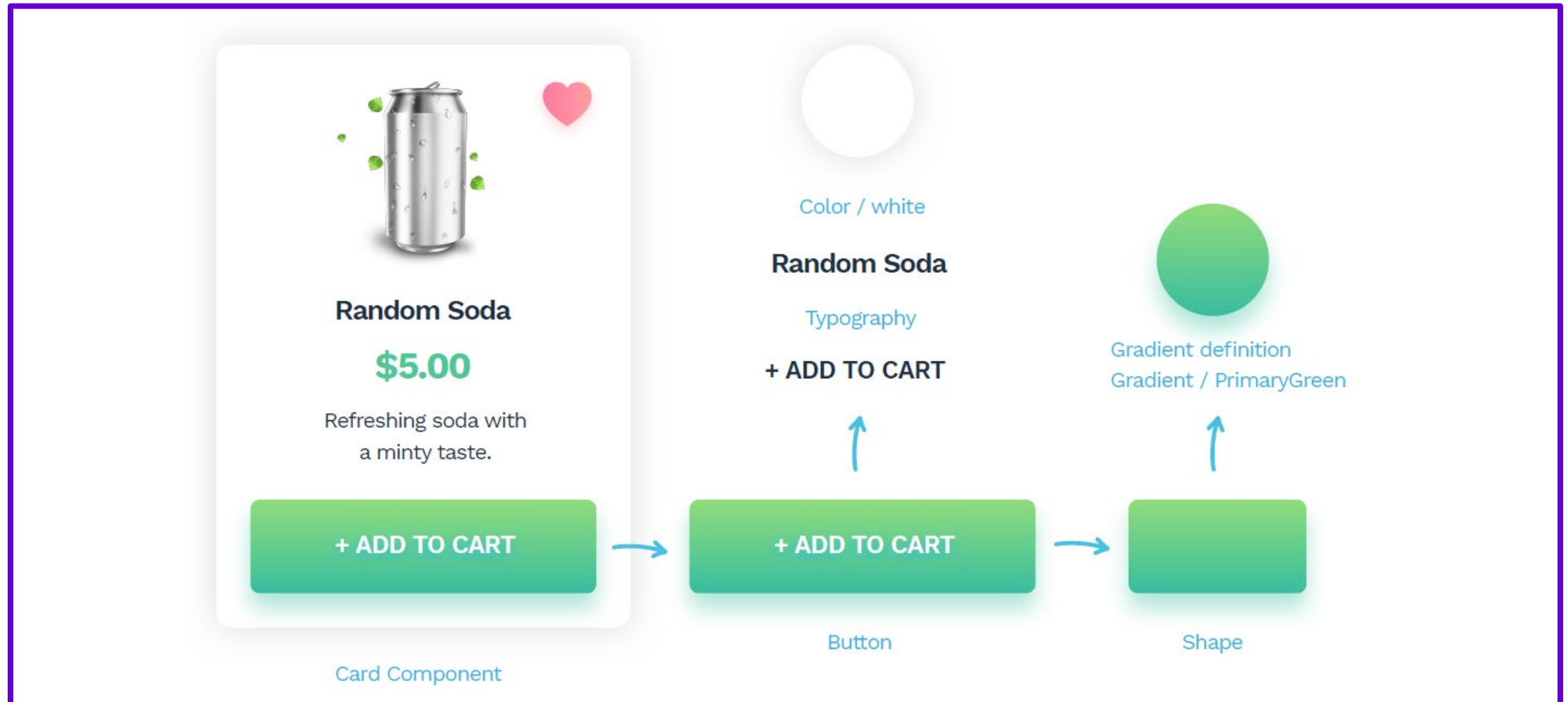
It seems obvious that the best approach here is to use a CMS, right? **No! it's best to build it out in code only**

It will help him understand the changes and rules a lot better, as he can ask specific questions when adding the changes to the wiki

VISUAL DOCUMENTATION

Each page of the system wiki should also have a place for visual documentation - mostly in the form of screenshots with short descriptions.

TREE-LIKE STRUCTURE



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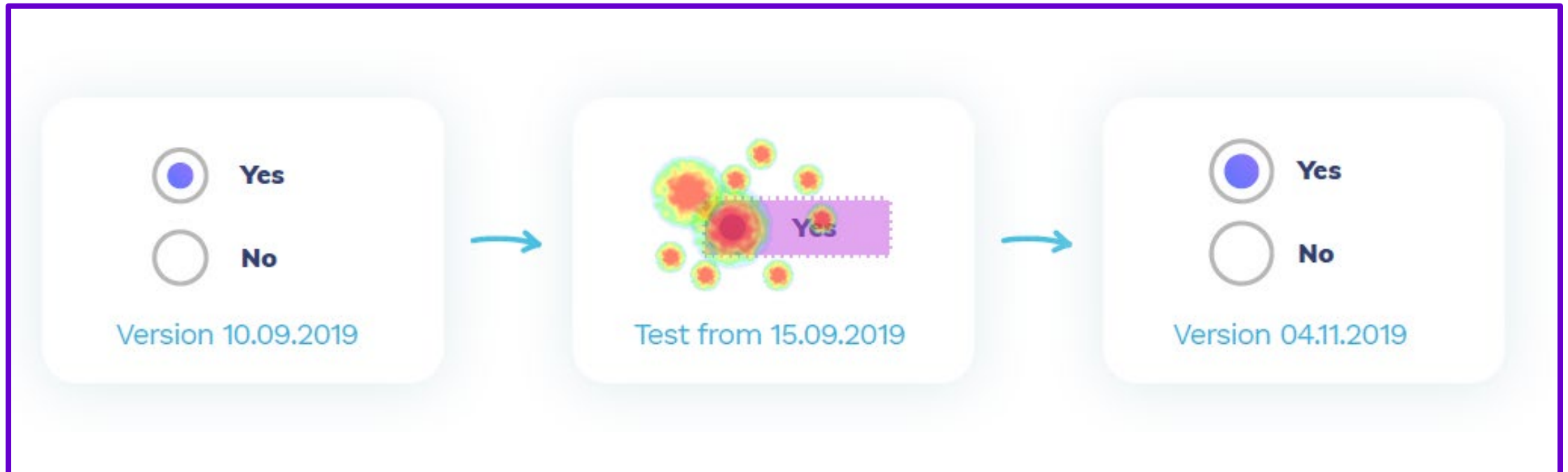
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RESEARCH AND DEVELOPMENT

When the descriptions, versioning, and history are all in, you can add the ability to document your research.



THE TEAM

To create a Design System team, you need at least:
a UI/UX designer ,a product owner, a front-end developer

The **best developers** for the job are the **ones who understand design** and have at least the **basic UI design knowledge already**. **Same with the designer** - it's best if they understand at least the principles behind CSS

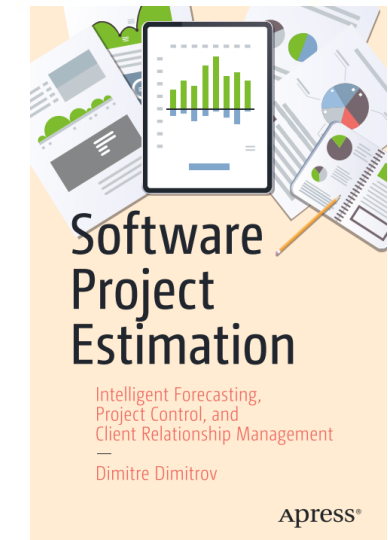
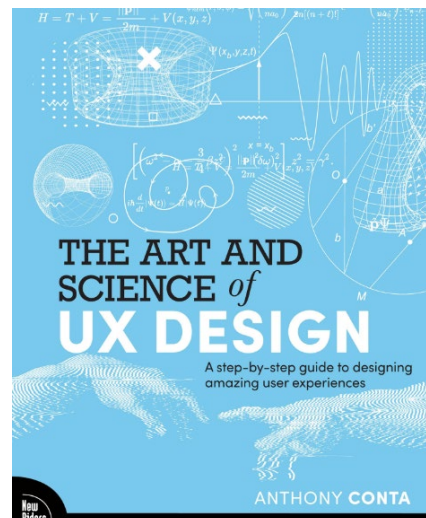
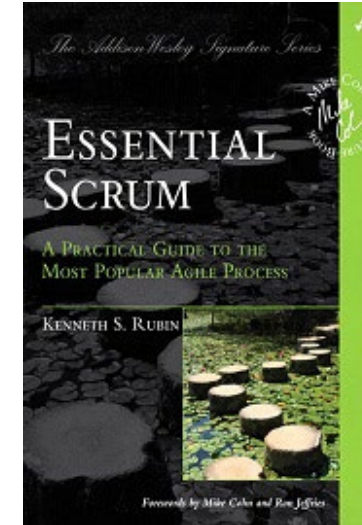
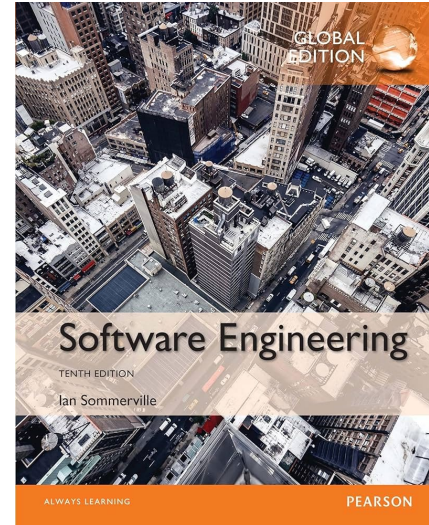
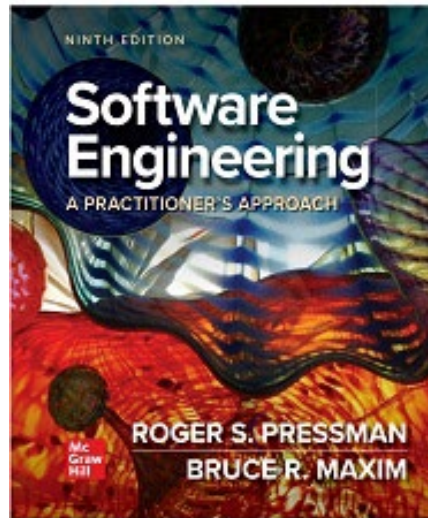
Systemic Approach

A Systemic Approach to components and screens is the future of design.

It allows you to focus on what's important (how things work) and not worry about the quality (as the components are pre-made and built correctly)

The Design System should be your baseline, but feel free to expand it and build additional components as long as they bring value to the project.

Course References



Course References

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- [2] Ockerman, S. and S. Reindl (2019). Mastering professional scrum: A practitioner's guide to overcoming challenges and maximizing the benefits of agility, Addison-Wesley Professional.
- [3] Martin, R. C. (2019). Clean Agile, Pearson Education.
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